What can waves tell us?

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish a signal's travel from the motion of its source.

SUCCESS CRITERIA - I can...

1. Identify what a wave can carry.
2. Separate signal travel time from source speed.
3. Name the measurements or details you need to explain an observation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "wave" and "signal" to explain your answer. Sentence starters are optional.

wave: a disturbance that can carry energy and information

signal: a disturbance that carries information

evidence: observations or measurements that support an idea

01 Warm-up: bring back an earlier idea

A sphere's radius becomes **3** times as large at fixed density. Derive the area-to-mass multiplier. **Explain why kinetic energy per** mass alone does not predict survival during entry.

Word help: derive = show the reasoning or algebra that produces a relationship

Sentence starter: I predict … . I used … to make this prediction because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: distance = speed × time
t = d/c for light in vacuum.
A signal takes time to travel. Receiving its light does not mean the source traveled with it.

Worked reasoning: A pulse travels **4 m** along a rope in **2 s**, so its average propagation speed is **2 m/s**. A marked rope segment moves locally; it does not ride **4 m** along with the pulse.

Your turn: Sketch three pulse positions and **mark the same rope segment** in each snapshot.

Sentence starter: In my work, I used … to show … . I labeled it … .

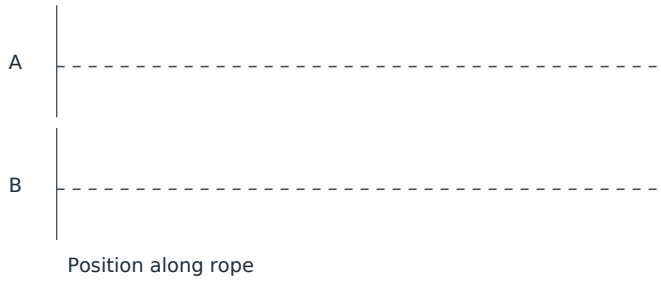


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A pulse travels along a rope while each marked rope segment moves mainly up and down. Does the material travel all the way with the pulse?

Sentence starter: I notice … . This supports … because … .

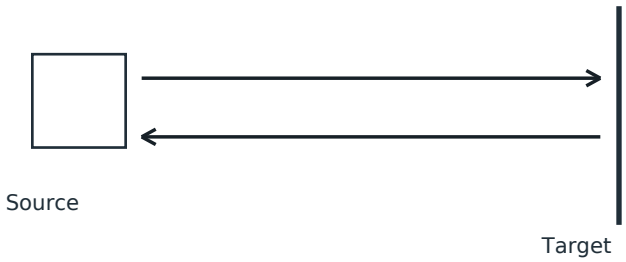


Marzano check - circle: ① ② ③

04 Apply the physics

A radar station receives an echo **0.020 s** after transmission.
1. Use c = **3.0 × 10⁸ m/s**.
2. **Label the outgoing** and returning paths, calculate target range and **explain the factor of 2**.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student says seeing an asteroid’s light tells us exactly where it is “right now.” Explain one limitation.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Two objects at **290 K** have different masses and materials. Must their internal energies be equal? **Give two reasons**.
Label temperatures, samples and relevant energy transfers.

Sentence starter: Both … . They differ in … because … .

Sample A

Sample B

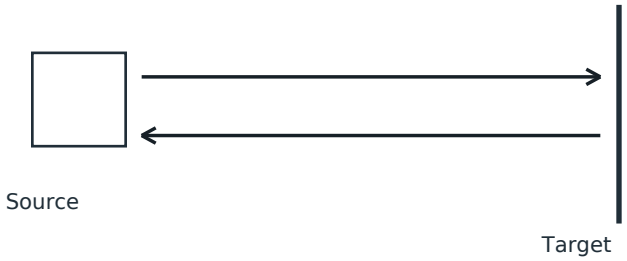
Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

07 Exit check

Two echoes return at **0.010** and **0.016 s** with the same wave speed. **Compare ranges** and **explain whether** echo delay alone gives mass.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A signal travels **300 m** in **0.9 s** with timing bounds **0.85**-**0.95 s**. **Find central speed** and bounds, treating distance as exact.

Sentence starter: I found … [unit]. This value means … in this situation.

Distance (m)	Time (s)	Speed (m/s)
___	___	___
___	___	___

Marzano check - circle: 1 2 3